

Paper A-08: Earth Pollution: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

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Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux Φ , constraint C , surface responsiveness S , and structural memory M_s . The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

Notation. In Part IV, C names the active constraint or capacity burden in the system being discussed. Φ is the driving flow, S is the surface response, and M_s is retained structural memory. Λ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

A healthy ecosystem is a fluid network operating in thermodynamic equilibrium. Natural waste products, such as fallen leaves or animal remains, represent chemical flux that is immediately processed and recycled by detritivores and microbial networks.

When humans introduce synthetic pollutants (e.g., microplastics, heavy metals, or persistent organic pollutants), they are injecting a massive, highly structured flux into an ecosystem that possesses absolutely no topological pathways to process it. In CDFD terms, pollution (maladaptive C accumulation) is an informational and energetic throughput that faces an infinite natural constraint.

3 The Bioremediation Adaptive Surface

We govern the environmental processing capacity using the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a contaminated ecological zone (e.g., a river basin or a soil horizon):

- **Flux** (Φ): The incoming rate of synthetic toxins and heavy metals.
- **Constraint** (C): The chemical resistance to degradation. Synthetic polymers possess extremely high chemical bonds that natural enzymes cannot easily break, acting as a massive biological friction limit.
- **Responsiveness** (S): The bioremediation capacity of the local microbial and fungal communities attempting to adapt and evolve to consume the novel flux.
- **Memory** (M_s): Bioaccumulation and environmental persistence. Because the toxins cannot be processed, they physically accumulate in the tissue of apex predators and the soil matrix. This permanently locks the system into a degraded, toxic state.

3.1 Bioaccumulation as Topological Ischemia

When natural organic flux enters the system, C is low, and S processes it efficiently ($\Psi_s \approx 1$).

When a synthetic pollutant enters, the local constraint C is effectively infinite. The biological network cannot degrade it. Consequently, the pollutant remains in the system. The physical presence of the toxin actively destroys the underlying microbiota and apex species, dropping the systemic responsiveness S toward zero.

Because the toxin persists, the systemic memory M_s becomes locked into a state of chronic poisoning. The entire network enters a severe bottleneck ($\Psi_s \ll 1$). The ecosystem ceases to

function as an active thermodynamic conduit and instead becomes a stagnant, necrotic sink for industrial waste.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Conclusion

pollution (maladaptive C accumulation) is the thermodynamic equivalent of a biological stroke. It is the injection of an anomalous flux into a network that lacks the topological architecture to process it. By framing environmental toxicity within the CDFD ontology, we frame the hypothesis that ecological collapse from pollution (maladaptive C accumulation) is the model-predicted failure mode of overwhelming the biosphere’s structural constraint limitations.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part’s `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

6 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.